

Mode 7 — Square-BFP (centered / opt)

DP16 (16×8) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → block exponents → DP8 (4×4) → centered aligned add-and-square

Step 1 — DP16 (16×8)

$$\text{DP16}(16 \times 8) = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP8 (16×8)

$$\text{DP16}(16 \times 8) = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8}(16 \times 8)} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8}(16 \times 8)}$$

Step 3 — DP8 (16×4)

$$\text{DP16}(16 \times 8) = 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8}(16 \times 4)} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8}(16 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i b_i^{hi}}_{\text{DP8}(16 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i^{lo}}_{\text{DP8}(16 \times 4)}$$

Step 4 — DP8 (8×4)

$$\begin{aligned} \text{DP16}(16 \times 8) = & 2^{12} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hi}}_{\text{DP8}(8 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{lo}}_{\text{DP8}(8 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hi}}_{\text{DP8}(8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{lo}}_{\text{DP8}(8 \times 4)} \\ & + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hi}}_{\text{DP8}(8 \times 4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{lo}}_{\text{DP8}(8 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hi}}_{\text{DP8}(8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{lo}}_{\text{DP8}(8 \times 4)} \end{aligned}$$

Step 5 — Block exponents (one exponent block per DP8 (8×4))

$$M = 2^{-\delta_0} \left(2^{12} \sum_{i=0}^7 a_i^{hi} b_i^{hi} + 2^8 \sum_{i=0}^7 a_i^{hi} b_i^{lo} + 2^4 \sum_{i=0}^7 a_i^{lo} b_i^{hi} + \sum_{i=0}^7 a_i^{lo} b_i^{lo} \right) + 2^{-\delta_1} \left(\dots i \in [8, 15] \dots \right)$$

$E_k = E_{a,k} + E_{b,k}$; $E = \max(E_0, E_1)$, $\delta_k = E - E_k$; truncating shifts, on block sums only; $\text{DP16}(16 \times 8) = M \cdot 2^E$.

Step 6 — DP8 (4×4)

$$\begin{aligned} M = & 2^{-\delta_0} \left(2^{16} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{hi}}_{\text{DP8}(4 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{hi}}_{\text{DP8}(4 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{lo}}_{\text{DP8}(4 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{lo}}_{\text{DP8}(4 \times 4)} \right. \\ & \left. + 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{hi}}_{\text{DP8}(4 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{hi}}_{\text{DP8}(4 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{lo}}_{\text{DP8}(4 \times 4)} + \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{lo}}_{\text{DP8}(4 \times 4)} \right) \\ & + 2^{-\delta_1} \left(2^{16} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{hi}}_{\text{DP8}(4 \times 4)} + 2^{12} \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{hi}}_{\text{DP8}(4 \times 4)} + 2^{12} \underbrace{\sum_{i=8}^{15} a_{H,i}^{hi} b_i^{lo}}_{\text{DP8}(4 \times 4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_{L,i}^{hi} b_i^{lo}}_{\text{DP8}(4 \times 4)} \right. \\ & \left. + 2^8 \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{hi}}_{\text{DP8}(4 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{hi}}_{\text{DP8}(4 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{H,i}^{lo} b_i^{lo}}_{\text{DP8}(4 \times 4)} + \underbrace{\sum_{i=8}^{15} a_{L,i}^{lo} b_i^{lo}}_{\text{DP8}(4 \times 4)} \right) \end{aligned}$$

Step 7 — Centered aligned add-and-square (subtract 8 from unsigned nibbles; per DP8 (4×4) add $c = 0/512/2048$ for $0/1/2$ unsigned nibbles; eight rows per K-half k , each weighted $2^{-\delta_k}$)

$$M = 2^{-\delta_0} 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hi})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hi})^2 \right)}_{\text{DP8}(4 \times 4)}$$

$$\begin{aligned}
& + 2^{-\delta_0} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + b_i^{hi})^2 - \sum_{i=0}^7 (a_{L,i}^{hi} - 8)^2 - \sum_{i=0}^7 (b_i^{hi} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_0} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{lo} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 - \sum_{i=0}^7 (b_i^{lo} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_0} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{lo} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 - \sum_{i=0}^7 (b_i^{lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_0} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + b_i^{hi})^2 - \sum_{i=0}^7 (a_{H,i}^{lo} - 8)^2 - \sum_{i=0}^7 (b_i^{hi} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_0} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + b_i^{hi})^2 - \sum_{i=0}^7 (a_{L,i}^{lo} - 8)^2 - \sum_{i=0}^7 (b_i^{hi} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_0} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{lo} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_0} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{lo} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_1} 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{hi})^2 - \sum_{i=8}^{15} (a_{H,i}^{hi})^2 - \sum_{i=8}^{15} (b_i^{hi})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_1} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + b_i^{hi})^2 - \sum_{i=8}^{15} (a_{L,i}^{hi} - 8)^2 - \sum_{i=8}^{15} (b_i^{hi} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_1} 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i}^{hi} + (b_i^{lo} - 8))^2 - \sum_{i=8}^{15} (a_{H,i}^{hi} - 8)^2 - \sum_{i=8}^{15} (b_i^{lo} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_1} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + (b_i^{lo} - 8))^2 - \sum_{i=8}^{15} (a_{L,i}^{hi} - 16)^2 - \sum_{i=8}^{15} (b_i^{lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_1} 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + b_i^{hi})^2 - \sum_{i=8}^{15} (a_{H,i}^{lo} - 8)^2 - \sum_{i=8}^{15} (b_i^{hi} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_1} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + b_i^{hi})^2 - \sum_{i=8}^{15} (a_{L,i}^{lo} - 8)^2 - \sum_{i=8}^{15} (b_i^{hi} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_1} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + (b_i^{lo} - 8))^2 - \sum_{i=8}^{15} (a_{H,i}^{lo} - 16)^2 - \sum_{i=8}^{15} (b_i^{lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{-\delta_1} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + (b_i^{lo} - 8))^2 - \sum_{i=8}^{15} (a_{L,i}^{lo} - 16)^2 - \sum_{i=8}^{15} (b_i^{lo} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

Step 8 — All components

$$\begin{aligned}
\text{PE} &= 2^{-\delta_0} \left(2^{16} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hi})^2 + 2^{12} \sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + b_i^{hi})^2 + 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{lo} - 8))^2 + 2^8 \sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{lo} - 8))^2 \right. \\
&\quad \left. + 2^8 \sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + b_i^{hi})^2 + 2^4 \sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + b_i^{hi})^2 + 2^4 \sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{lo} - 8))^2 + \sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{lo} - 8))^2 \right) \\
&\quad + 2^{-\delta_1} \left(2^{16} \sum_{i=8}^{15} (a_{H,i}^{hi} + b_i^{hi})^2 + 2^{12} \sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + b_i^{hi})^2 + 2^{12} \sum_{i=8}^{15} (a_{H,i}^{hi} + (b_i^{lo} - 8))^2 + 2^8 \sum_{i=8}^{15} ((a_{L,i}^{hi} - 8) + (b_i^{lo} - 8))^2 \right. \\
&\quad \left. + 2^8 \sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} ((a_{H,i}^{lo} - 8) + (b_i^{lo} - 8))^2 + \sum_{i=8}^{15} ((a_{L,i}^{lo} - 8) + (b_i^{lo} - 8))^2 \right) \\
\alpha &= 2^{-\delta_0} \left(2^{16} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{hi} - 8)^2 + 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 \right. \\
&\quad \left. + 2^8 \sum_{i=0}^7 (a_{H,i}^{lo} - 8)^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{lo} - 8)^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 + \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 \right) \\
&\quad + 2^{-\delta_1} \left(2^{16} \sum_{i=8}^{15} (a_{H,i}^{hi})^2 + 2^{12} \sum_{i=8}^{15} (a_{L,i}^{hi} - 8)^2 + 2^{12} \sum_{i=8}^{15} (a_{H,i}^{hi} - 8)^2 + 2^8 \sum_{i=8}^{15} (a_{L,i}^{hi} - 16)^2 \right. \\
&\quad \left. + 2^8 \sum_{i=8}^{15} (a_{H,i}^{lo} - 8)^2 + 2^4 \sum_{i=8}^{15} (a_{L,i}^{lo} - 8)^2 + 2^4 \sum_{i=8}^{15} (a_{H,i}^{lo} - 16)^2 + \sum_{i=8}^{15} (a_{L,i}^{lo} - 16)^2 \right) \\
\beta &= 2^{-\delta_0} \left(2^{16} \sum_{i=0}^7 (b_i^{hi})^2 + 2^{12} \sum_{i=0}^7 (b_i^{hi} - 8)^2 + 2^{12} \sum_{i=0}^7 (b_i^{lo} - 8)^2 + 2^8 \sum_{i=0}^7 (b_i^{lo} - 16)^2 \right. \\
&\quad \left. + 2^8 \sum_{i=0}^7 (b_i^{hi} - 8)^2 + 2^4 \sum_{i=0}^7 (b_i^{hi} - 8)^2 + 2^4 \sum_{i=0}^7 (b_i^{lo} - 16)^2 + \sum_{i=0}^7 (b_i^{lo} - 16)^2 \right) \\
&\quad + 2^{-\delta_1} \left(2^{16} \sum_{i=8}^{15} (b_i^{hi})^2 + 2^{12} \sum_{i=8}^{15} (b_i^{hi} - 8)^2 + 2^{12} \sum_{i=8}^{15} (b_i^{lo} - 8)^2 + 2^8 \sum_{i=8}^{15} (b_i^{lo} - 16)^2 \right. \\
&\quad \left. + 2^8 \sum_{i=8}^{15} (b_i^{hi} - 8)^2 + 2^4 \sum_{i=8}^{15} (b_i^{hi} - 8)^2 + 2^4 \sum_{i=8}^{15} (b_i^{lo} - 16)^2 + \sum_{i=8}^{15} (b_i^{lo} - 16)^2 \right) \\
C &= (2^{-\delta_0} + 2^{-\delta_1}) (0 + 2097152 + 2097152 + 524288 + 131072 + 8192 + 32768 + 2048) = (2^{-\delta_0} + 2^{-\delta_1}) \cdot 4892672 \\
M &= \frac{1}{2} (\text{PE} - \alpha - \beta + C), \quad E = \max(E_0, E_1)
\end{aligned}$$